

LAB 3 – OOP

1. Define simple class.

```
struct Student
{
    public string name;
    public int age;
}
class Program
{
    static void Main(string[] args)
    {
        Student s1;
        s1.name = "John";
        s1.age = 20;
        Console.WriteLine("Student details: ");
        Console.WriteLine($"name: {s1.name}");
        Console.WriteLine($"age: {s1.age}");

    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3$ dotnet run
Student details:
name: John
age: 20
```

2. Add a method that prints data.

```
struct Student
{
    public string name;
    public int age;

    public void display()
    {
        Console.WriteLine("Student Details");
        Console.WriteLine($"Name: {name} & Age: {age}");
    }
}
class Program
{
    static void Main(string[] args)
    {
        Student s1;
        s1.name = "John Dawson";
        s1.age = 23;
        s1.display();
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3$ dotnet run
Student Details
Name: John Dawson & Age: 23
```

3. Class with a calculation method.

```

struct Rectangle
{
    public double Width;
    public double Height;

    public double calculateArea()
    {
        return Width * Height;
    }
}
class Program
{
    static void Main(string[] args)
    {
        Rectangle r1;
        r1.Height = 20;
        r1.Width = 42;
        Console.WriteLine($"Area of rectangle is {r1.calculateArea()}");
    }
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3$ dotnet run
Area of rectangle is 840

```

4. Method with parameters

```

struct Rectangle
{
    public double calculateArea(double Width, double Height)
    {
        return Width * Height;
    }
}
class Program
{
    static void Main(string[] args)
    {
        Rectangle r1;
        Console.WriteLine($"Area of rectangle is {r1.calculateArea(34, 23)}");
    }
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3$ dotnet run
Area of rectangle is 782

```

5. Multiple objects from same class

```

struct Student
{
    public string name;
    public int age;

    public void displayInfo()
    {
        Console.WriteLine("Student Details : ");
        Console.WriteLine($"Name: {name}");
        Console.WriteLine($"Age: {age}");
    }
}

```

```

    }
    class Program
    {
        static void Main(string[] args)
        {
            Student s1, s2;
            s1.name = "Jack Rider";
            s1.age = 23;
            s1.displayInfo();
            Console.WriteLine();
            s2.name = "Rosie Dawson";
            s2.age = 21;
            s2.displayInfo();
        }
    }
}

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3$ dotnet run
Student Details :
Name: Jack Rider
Age: 23

Student Details :
Name: Rosie Dawson
Age: 21

```

6. Simple behavior method

```

class Dog
{
    private string name;
    public Dog(string name)
    {
        this.name = name;
    }
    public void displayInfo()
    {
        Console.WriteLine($"{name} is barking !");
    }
}

class Program
{
    static void Main(string[] args)
    {
        Dog dog = new Dog("Rex");
        dog.displayInfo();
    }
}

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3$ dotnet run
Rex is barking !

```

7. Bank Account deposit and withdraw

```

class Bank
{
    private string accountNumber;
    private double balance;

    public Bank(string accountNumber, double balance)
    {
        this.accountNumber = accountNumber;
        this.balance = balance;
    }

    public void deposit(double amount)
    {
        balance += amount;
    }
    public void withdraw(double amount)
    {
        balance -= amount;
    }
    public void displayInfo()
    {
        Console.WriteLine("Account Number: " + accountNumber);
        Console.WriteLine("Balance: " + balance);
        Console.WriteLine();
    }
}

```

```

class Program
{
    static void Main(string[] args)
    {
        Bank acc1 = new Bank("062123X", 1000);
        acc1.deposit(2500);
        acc1.displayInfo();
        acc1.withdraw(500);
        acc1.displayInfo();
    }
}

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3\$ dotnet run

Account Number: 062123X

Balance: 3500

Account Number: 062123X

Balance: 3000

8. Book details and summary

```

class Book
{
    private string title;
    private string author;
    private int pages;
}

```

```
public Book(string title, string author, int pages)
{
    this.title = title;
    this.author = author;
    this.pages = pages;
}
public void PrintSummary()
{
    Console.WriteLine($"Title: {title}, Author: {author}, Pages: {pages}");
}
}
```

```
class Program
{
    static void Main(string[] args)
    {
        Book b1 = new Book("The Hobbit", "Rayan Dawson", 300);
        Book b2 = new Book("The Lord of the Rings", "Harry Mark", 700);
        b1.PrintSummary();
        b2.PrintSummary();
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab3$ dotnet run
Title: The Hobbit, Author: Rayan Dawson, Pages: 300
Title: The Lord of the Rings, Author: Harry Mark, Pages: 700
```